# CONVERGENCE OF COLLISION FLUCTUATION SUMS

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Abstract. The polarity field Dirichlet series $F^\circ(s) = \sum_p S^\circ(p)/p^s$, where $S^\circ$ is the centered collision invariant, converges at $s = 1$ by the collision PNT [4]. We prove that GRH implies convergence of $F^\circ$ for $\mathrm{Re}(s) > 1/2$, and conversely that any unshared off-line zero obstructs convergence. Under the hypothesis that off-line zeros are unshared (expected by the Grand Simplicity Hypothesis; no shared zeros observed for characters modulo 100), the two conditions are equivalent.

The avoidance deficit $D(\sigma) > 0$ measures anti-correlation between collision weights and prime character sums: characters that interact strongly with the digit function's bin partition interact weakly with the prime distribution, and vice versa. The avoidance reduces to a single inequality in $\mathbb{Q}(\sqrt{5})$: $6(r_{26} - r_{48}) + 35q_{26} + 25q_{48} < 0$, verified with $2 \times 10^8$ primes at 33 values of $\sigma$ in $[0.50, 1.14]$.

The convergence reduces to complement pairs: $F^\circ = \sum S^\circ(a) \cdot \Delta(s, a)$. Computation shows 99.9% cancellation at $\sigma = 1/2$, with gross energy growing $506\times$ while the net grows only $11\times$. The convergence of $F^\circ$ for $\mathrm{Re}(s) > 1/2$ is conjectured; the gap between the proved $O(x \exp(-c\sqrt{\log x}))$ and the needed $O(\sqrt{x})$ is the distance from the classical zero-free region to GRH.

## 1. Introduction

The digit function $\lfloor br/p \rfloor$ partitions residues additively (by the ratio $r/p$). The prime distribution is multiplicative. The collision count measures the interaction between these two structures. The centered collision deviation $S^\circ(p) = \sum_\chi \widehat{c}(\chi)\, \chi(p)$ decomposes into odd characters modulo $m = b^2$ (primitive odd for prime bases) [2, 3]. Paper D [4] proved the collision PNT, the $L$-value extraction, the diagonal–Bernoulli identity, and the $\mathbb{Q}(\sqrt{5})$ weight decomposition.

This paper asks what the collision invariant knows about $L$-function zeros.

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The polarity field Dirichlet series $F^\circ(s)$ converges at $s = 1$ by the collision PNT. How far into the critical strip does it penetrate? The answer (Section 2): as far as the zeros allow. GRH implies convergence for $\mathrm{Re}(s) > 1/2$, and an unshared off-line zero obstructs it. The collision weights are anti-correlated with the prime character sums (Section 3): characters that see the digit function strongly see the primes weakly. This avoidance reduces to one inequality in $\mathbb{Q}(\sqrt{5})$ (Section 5). The complement pair structure produces 99.9% cancellation at $\sigma = 1/2$ (Section 4), driven by the centering of the periodic table (Section 6). Section 7 states the convergence conjecture and the distance that remains.

## 2. The Convergence Equivalence

**Definition 2.1.** The *polarity field Dirichlet series* is

$$F^\circ(s) = \sum_{\substack{p>b \\ p\nmid b}} \frac{S^\circ(p)}{p^s},$$

where $\mathcal{X}$ denotes the odd characters surviving after centering (primitive odd for prime bases; see [4]). In any zero-free region of all $L(s,\chi)$ with $\chi \in \mathcal{X}$, the Fourier decomposition gives the evaluation $F^\circ(s) = \sum_{\chi\in\mathcal{X}} \widehat{c}(\chi)[\log L(s,\chi) - R(s,\chi)]$, where $R(s,\chi) = \sum_p \sum_{k\geq 2} \chi(p)^k/(k\,p^{ks})$ converges absolutely for $\mathrm{Re}(s) > 1/2$.

**Theorem 2.2** (Convergence equivalence)**.** *Assume $\widehat{c}(\chi) \neq 0$ for all $\chi \in \mathcal{X}$, and that any zero $\rho$ of an $L(s,\chi)$ with $\mathrm{Re}(\rho) > 1/2$ is unshared: it is not simultaneously a zero of any other $L(s,\chi')$ in $\mathcal{X}$ (see Remark 2.3). Then:*

$$F^\circ(s) \text{ converges for all } \mathrm{Re}(s) > \tfrac{1}{2} \quad \Longleftrightarrow \quad \text{GRH for all } L(s,\chi),\ \chi \in \mathcal{X}.$$

*Proof.* ($\Leftarrow$) Under GRH, the explicit formula gives $\sum_{p\leq x} \chi(p) = O(\sqrt{x}\,\log^2 x)$ for each $\chi \in \mathcal{X}$. Then $A(x) = \sum_{p\leq x} S^\circ(p) = \sum_\chi \widehat{c}(\chi) \sum_{p\leq x} \chi(p) = O(\sqrt{x}\,\log^2 x)$, and partial summation gives convergence of $F^\circ(\sigma)$ for real $\sigma > 1/2$. By the standard half-plane convergence theorem for Dirichlet series [6], convergence on the real line implies convergence (and analyticity) in the full half-plane $\mathrm{Re}(s) > 1/2$.

($\Rightarrow$) Suppose $L(\rho,\chi_0) = 0$ for some $\rho$ with $\mathrm{Re}(\rho) > 1/2$ and some $\chi_0 \in \mathcal{X}$, and suppose no other $L(s,\chi)$ vanishes at $\rho$. Then $\log L(s,\chi_0)$ has a logarithmic branch point at $\rho$ (locally, $\log L(s,\chi_0) = \nu \log(s-\rho) + \text{analytic}$, $\nu \geq 1$). All other terms are analytic near $\rho$. The sum $F^\circ$ inherits the branch point. But a Dirichlet series converging for all $\sigma > 1/2$ defines an analytic function in $\mathrm{Re}(s) > 1/2$ (a standard

property of Dirichlet series [5]), which cannot have a branch point at $\rho$ with $\mathrm{Re}(\rho) > 1/2$. Therefore $F^\circ$ does not converge for all $\sigma > 1/2$. $\square$

*Remark* 2.3. The condition $\widehat{c}(\chi) \neq 0$ follows from $L(1,\chi) \neq 0$ (Dirichlet) and $S_G(\chi) \neq 0$ (computed for all bases through $b = 60$), via the decomposition $\widehat{c} = -B_{1,\overline{\chi}} \cdot \overline{S_G}/\phi$.

The converse uses the hypothesis that off-line zeros are unshared (not simultaneously zeros of distinct $L$-functions in the decomposition). Shared zeros of distinct primitive $L$-functions are expected not to occur (the Grand Simplicity Hypothesis) and have not been observed computationally for any characters modulo 100.

## 3. The Avoidance Deficit

**Definition 3.1.** The *polarity field* on $(\mathbb{Z}/m\mathbb{Z})^*$ is $F(a) = S^\circ(a)$, the centered collision invariant. It is antisymmetric: $F(a) + F(m-a) = 0$.

**Definition 3.2.** The *prime character sum* is $P(s,\chi) = \sum_{p>b,\, p\nmid b} \chi(p)/p^s$ and the *avoidance deficit* is

$$D(s) = 1 - \frac{\sum_\chi |\widehat{c}(\chi)|^2 \, |P(s,\chi)|^2}{\left(\sum_\chi |\widehat{c}(\chi)|^2\right)\left(\frac{1}{N}\sum_\chi |P(s,\chi)|^2\right)},$$

where $N = |\mathcal{X}|$. When $D(s) > 0$, the collision weights concentrate on characters with small prime sums: the digit function avoids the characters that carry the most prime-level information.

**Observation 3.3** (Avoidance at $s = 1$)**.** For the 8 primitive odd characters modulo 100, computation with $10^7$ primes gives $D(1) = 0.1899 \pm 0.0009$ (the error is twice the maximum change from $5 \times 10^6$ to $10^7$ primes). The rank reversal is complete:

| $k$ | $\lvert\widehat{c}(\chi_k)\rvert^2$ | $\lvert P(1,\chi_k)\rvert^2$ |
|---|---|---|
| $2, 18$ | 0.2133 | 0.0161 |
| $4, 16$ | 0.1309 | 0.0343 |
| $6, 14$ | 0.0567 | 0.0168 |
| $8, 12$ | 0.0191 | 0.0479 |

The largest collision weight pair carries the smallest prime sum, and vice versa.

**Observation 3.4** (Universality and monotone growth)**.** $D(1) > 0$ for every tested base ($b = 5, 7, 10, 11, 13$). In base 10, $D(\sigma)$ grows monotonically from $D(2) = -0.14$ through $D(1) = +0.19$ to $D(1/2) = +0.55$: the avoidance is maximal at the fixed line of the functional equation.

Is the avoidance a consequence of antisymmetry alone, or of the digit function's specific geometry?

**Observation 3.5** (Five-class experiment)**.** Five classes of tables on $(\mathbb{Z}/100\mathbb{Z})^*$, 20 random instances each, $2 \times 10^6$ primes:

| Table class | $D(1) > 0$ | Interpretation |
|---|---|---|
| Real periodic table | 1/1 | always avoidance |
| Random antisymmetric | 11/20 | coin flip |
| Shuffled (same values) | 14/20 | coin flip |
| Fully random | 6/20 | below chance |
| Symmetric | 0/20 | zero weights |

The qualitative pattern is clear: symmetric tables give zero weights; random antisymmetric tables give avoidance roughly half the time; only the real table gives avoidance consistently. Antisymmetry is necessary (it creates nonzero weights on odd characters) but not sufficient (the specific assignment of collision counts to classes, determined by $\lfloor br/p \rfloor$, is essential).

## 4. The Pair Decomposition

The antisymmetry $S^\circ(a) = -S^\circ(m-a)$ pairs the coprime classes. Each pair contributes to $F^\circ$ through the Chebyshev bias between its two classes.

**Proposition 4.1** (Pair decomposition)**.** $F^\circ(\sigma) = \sum_{a<m/2,\, \gcd(a,m)=1} S^\circ(a) \cdot \Delta(\sigma, a)$, *where* $\Delta(\sigma, a) = \sum_{p\equiv a} p^{-\sigma} - \sum_{p\equiv m-a} p^{-\sigma}$.

*Proof.* Group by complement pairs and use $S^\circ(m-a) = -S^\circ(a)$. □

**Proposition 4.2** ($F^\circ$ is real)**.** *For real* $\sigma$, $F^\circ(\sigma) \in \mathbb{R}$.

*Proof.* Conjugate characters contribute conjugate terms. □

**Observation 4.3** (Polarity–bias anti-correlation)**.** Pairs with large polarity carry small bias. The dominant pair $(9, 91)$ has $|S^\circ| = 7.3$ and $|\Delta(1)| = 0.007$. All pair products are bounded by 0.07.

**Observation 4.4** (Cancellation)**.** The fraction of gross pair-product energy that cancels in the signed sum improves monotonically:

| $\sigma$ | cancellation | $\lvert F^\circ(\sigma)\rvert$ |
|---|---|---|
| 1.0 | 95.8% | 0.14 |
| 0.8 | 98.8% | 0.31 |
| 0.6 | 99.8% | 0.75 |
| 0.5 | 99.9% | 1.54 |

The individual pair products grow explosively: $506\times$ from $\sigma = 1$ to $\sigma = 1/2$. But the signed sum grows only $11\times$. The polarity field cancels its own growth as it approaches the critical line.

## 5. The $\mathbb{Q}(\sqrt{5})$ Reduction

The collision weights live in $\mathbb{Q}(\sqrt{5})$ (Theorem 6.1 of [4]): the squared diagonal sums have the form $|S_G(\chi_k)|^2 = A_k + 2\epsilon_k\sqrt{5}$, where $\epsilon_k = \pm 1$ distinguishes complement characters. The $\sqrt{5}$ arises from the 5th roots of unity in the character theory of $(\mathbb{Z}/25\mathbb{Z})^*$.

**Theorem 5.1** (The $\mathbb{Q}(\sqrt{5})$ reduction)**.** *With $r_{26} = (p_2 + p_6)/2$, $r_{48} = (p_4 + p_8)/2$ (the rational averages) and $q_{26} = (p_2 - p_6)/(2\sqrt{5})$, $q_{48} = (p_4 - p_8)/(2\sqrt{5})$ (the $\sqrt{5}$ components), where $p_k = |P(\sigma, \chi_k)|^2$:*

$$D(\sigma) > 0 \quad \Longleftrightarrow \quad 6(r_{26} - r_{48}) + 35\, q_{26} + 25\, q_{48} < 0.$$

*Proof.* Write $w_k = w_r + w_q\sqrt{5}$ for the collision weight $|\widehat{c}(\chi_k)|^2$, where $w_r = (A_k^2 + 20)/\varphi^2$ and $w_q = 4A_k\epsilon_k/\varphi^2$ ($\varphi = \varphi(m) = 40$). For a complement pair $(k, k')$: $w_k p_k + w_{k'} p_{k'} = 2(w_r r + 5w_q q)$, which is rational. The deficit $D = 1 - \sum w_i p_i / (\sum w_i \cdot \bar{p})$ is positive iff $\sum w_i p_i < \sum w_i \cdot \bar{p}$, where $\bar{p} = (r_{26} + r_{48})/2$.

Expanding both sides: $\sum w_i p_i$ has coefficients $4w_{2,r}$ on $r_{26}$ and $4w_{4,r}$ on $r_{48}$ plus $20w_{2,q}$ on $q_{26}$ and $20w_{4,q}$ on $q_{48}$. The product $\sum w_i \cdot \bar{p}$ has coefficient $\sum w_i/2$ on both $r_{26}$ and $r_{48}$. With $w_{2,r} = 216/1600$, $w_{4,r} = 120/1600$, $\sum w_i = 1344/1600$: the $r_{26}$ residual is $4 \cdot 216 - 672 = 192$ and the $r_{48}$ residual is $4 \cdot 120 - 672 = -192$. The $q$ coefficients are $20 \cdot 56 = 1120$ and $20 \cdot 40 = 800$. Dividing by 32: $6(r_{26} - r_{48}) + 35q_{26} + 25q_{48} < 0$. □

The avoidance has two sources. The rational term $6(r_{26} - r_{48})$ measures the imbalance in average prime sums between the two complement pairs. The $\sqrt{5}$ term $35q_{26} + 25q_{48}$ measures the imbalance within each pair: characters carrying $+\sqrt{5}$ in their collision weights see smaller prime sums than those carrying $-\sqrt{5}$. The digit function's bin geometry and the prime distribution pull in opposite directions, separated by $\sqrt{5}$.

**Observation 5.2** (Computational verification)**.** With $2 \times 10^8$ primes, tracked across 7 doublings of the prime limit. The error estimate is twice the maximum change across the last three doublings (a convergence diagnostic, not a rigorous tail bound):

| $\sigma$ | value | error | status |
|---|---|---|---|
| 1.00 | $-0.073$ | 0.000 | numerically verified |
| 0.80 | $-0.555$ | 0.006 | numerically verified |
| 0.60 | $-2.898$ | 0.447 | numerically verified |
| 0.50 | $-6.034$ | 3.777 | verified with margin |

The inequality strengthens monotonically across 33 tested values of $\sigma$ in $[0.50, 1.14]$. For base $b = 2p$ with $p$ an odd prime, the same structure gives an avoidance inequality in $\mathbb{Q}(\sqrt{p})$.

## 6. Why the Cancellation Improves

**Proposition 6.1** (Centering [2])**.** *Within each spectral class $R \in \{0, 2, 6, 8\}$: $\sum_{a:(a-1) \bmod 10=R} S^\circ(a) = 0$.*

*Proof.* $S^\circ(a) = \mathcal{T}(a) - \overline{S}_R$ where $\overline{S}_R$ is the class mean. The sum within each class telescopes to zero by construction. □

The centering is exact and independent of $\sigma$: it comes from the periodic table, not from the primes. The primes contribute the pair biases, which by Dirichlet equidistribution have the same leading term for all pairs. The centering kills that leading term. As $\sigma$ decreases, more primes enter each pair bias, the biases become more uniform, and the centering cancels a larger fraction of the total. The centering is algebraic; the uniformity is arithmetic. Together they turn 95.8% into 99.9%.

## 7. The Convergence Target

**Conjecture 7.1** (Polarity field convergence)**.** *For each base $b \geq 3$ and each $\sigma > 1/2$: $F^\circ(\sigma)$ converges.*

By Theorem 2.2, GRH implies this conjecture, and (under the non-degeneracy hypothesis of Remark 2.3) the converse holds.

The partial sum $A(x) = \sum_{p \leq x} S^\circ(p)$ grows as $O(\sqrt{x})$ empirically through $5 \times 10^6$ primes, with anti-persistent autocorrelation (the Lemke Oliver–Soundararajan effect [7]). By partial summation, $O(\sqrt{x})$ growth implies convergence for $\sigma > 1/2$. The collision PNT gives $A(x) = O(x \exp(-c\sqrt{\log x}))$; the gap between this and $O(\sqrt{x})$ is the distance from the classical zero-free region to GRH.

## 8. Remarks

**What the avoidance means.** The collision weights concentrate on characters where $|L(1, \chi)|$ is large. The avoidance says these same

characters have small $|P(s,\chi)|$ in the strip. Since $|P|$ is large when $L$ has zeros nearby, the collision invariant avoids the characters whose $L$-functions have zeros close to the evaluation point. Under GRH, all zeros lie on $\mathrm{Re}(s) = 1/2$, and the avoidance is maximal there. The monotone growth of $D(\sigma)$ toward $\sigma = 1/2$ is consistent with this picture.

**Open questions.**

(1) Prove the avoidance inequality (Theorem 5.1) analytically for all $\sigma > 1/2$.
(2) Prove $A(x) = O(\sqrt{x})$ from the arithmetic of $\lfloor br/p \rfloor$.
(3) Extend the $\mathbb{Q}(\sqrt{p})$ avoidance to all bases $b = 2p$.

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